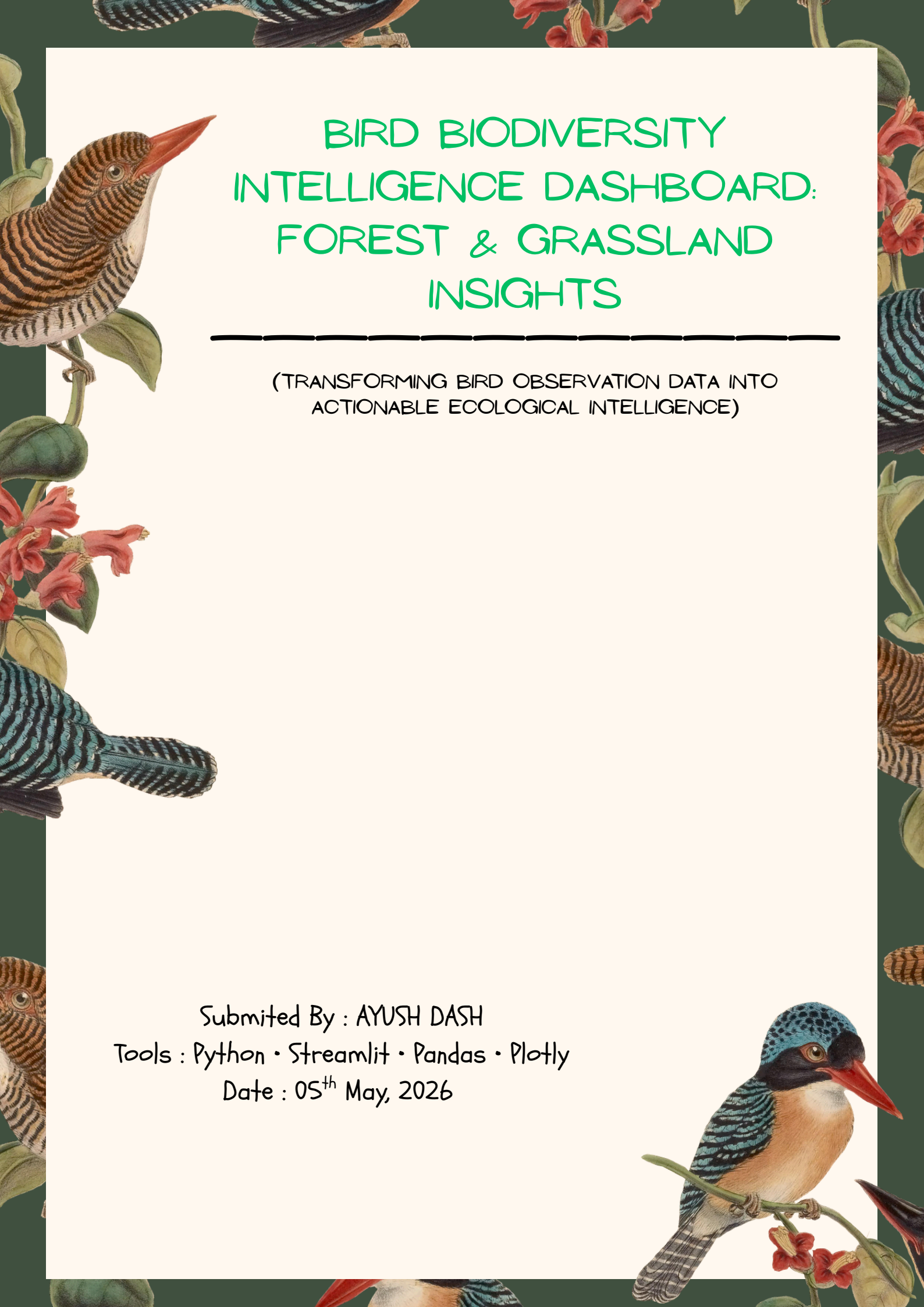


The page features a decorative border on the left and right sides. On the left, there are illustrations of a brown and white striped bird perched on a branch, and a blue and white striped bird. On the right, there are illustrations of a blue and white striped bird and a brown and white striped bird. Red flowers are also part of the border.

BIRD BIODIVERSITY INTELLIGENCE DASHBOARD: FOREST & GRASSLAND INSIGHTS

(TRANSFORMING BIRD OBSERVATION DATA INTO
ACTIONABLE ECOLOGICAL INTELLIGENCE)

Submitted By : AYUSH DASH
Tools : Python • Streamlit • Pandas • Plotly
Date : 05th May, 2026



EXECUTIVE SUMMARY

1. Executive Summary

This project provides a data-driven study of biodiversity for birds based on observational data gathered from forest and grassland ecosystems. The aim of this project is the creation of an interactive dashboard, which could convert raw ecological data into valuable insight through the application of data analytics technologies.

In total, the used dataset contains more than 17,000 observations from various parks, species, and different environmental conditions. It should be mentioned that there are strong seasonal trends as birds are most active in the period of monsoons. This fact indicates migration and reproduction periods of the birds under consideration.

It has been discovered that species diversity is higher in forest habitats while observations are mostly made in grasslands. From the perspective of spatial analysis, biodiversity hot spots have been revealed, in particular, ANTI Park, which contains the largest share of observations.

Another notable finding relates to a skewed distribution of the species in question with several species making up the majority of observations. Finally, environmental factors such as temperature, humidity, and sky condition have significant impact on birds.

OBJECTIVE & DATASET

2. Objective of the Project

The main goals of the proposed project are as follows:

- Analysis of birds' biodiversity in forests and grasslands
- Identification of time trends (seasonal and annual trends)
- Assessment of the environment's influence on birds' biodiversity
- Detection of biodiversity hotspots
- Development of an interactive dashboard

3. Dataset Overview

Data Source Files:-

File Name	Description
Bird_Monitoring_Data_FOREST.xlsx	Forest ecosystem bird observations
Bird_Monitoring_Data_GRASSLAND.xlsx	Grassland ecosystem bird observations

Dataset Summary:-

Feature	Details
Total Records	17,000+ observations
Data Format	Excel (.xlsx)
Ecosystems	Forest & Grassland
Time Coverage	Multi-year
Locations Covered	11 Parks

Key Variables:-

Column Name	Description
Common_Name	Bird species (common name)
Scientific_Name	Taxonomic classification
Habitat	Forest / Grassland
Park	Observation location
Date / Year	Time dimension
Season	Derived seasonal category
Temperature	Weather condition
Humidity	Atmospheric condition
Wind	Wind speed/category
Sky	Sky visibility condition
Distance	Observation distance

DATA PROCESSING & DASHBOARD

4. Data Cleaning & Preparation

Data Cleaning:-

- Duplicates removed
- Missing values managed via filtering
- Categorical variables standardized
- Inconsistent data cleaned out

Data Transformation:-

- Year and Month extracted from date
- Season created as a new variable
- Grouping and aggregation through GroupBy
- Creation of biodiversity indicators

5. Dashboard Overview

Tools Used:-

- Python (Pandas)
- Plotly (Visualization)
- Streamlit (Dashboard UI)

Dashboard Modules:-

Module	Description
Temporal	Seasonal & yearly trends
Spatial	Location-based biodiversity
Species	Species distribution & dominance
Environment	Weather impact analysis
Map	Geographic visualization
SQL	Interactive data exploration

Filters

Habitat

- Forest
- Grassland

Park

- CANYON
- CATD
- CHERRY
- GROVE
- MOUNTAIN
- RIVER
- SAGE
- VALLEY

Year

- 2018

Bird Biodiversity Intelligence Dashboard:
Forest & Grassland Insights

Turning Bird Observation Data into Actionable Insights for Conservation and Ecosystem Management.

Observations
17077

Species
126

Parks
11

Habitats
2

Temporal

Spatial

Species

Environment

Map

SQL

Monthly Bird Activity Trend

Monthly Bird Observation Pattern

Count

MayJunJulMonth, NameAug

Seasonal Distribution

Bird Observations Across Seasons

Count

MonsoonSummer

Yearly Trend (Smoothed)

Yearly Bird Observations

Count

Trend (Smoothed)

20172017.520182018.52019Year

Peak Activity Month

Peak Bird Activity by Month

Count

MayJunJulMonth

Activity Heatmap (Year vs Month)

Bird Activity Intensity

Sum of Count

3.544.555.566.5777.5Month

Smart Insights

• Peak Season: Monsoon

• Top Park: ANTI

• Top Species: Northern Cardinal

Download Data

Bird Biodiversity Intelligence Dashboard: Forest & Grassland Insights

Turning Bird Observation Data into Actionable Insights for Conservation and Ecosystem Management

Observations
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Spatial

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Environment

Map

SQL

Spatial Biodiversity Insights

Observation Distribution by Location Type

Location_Type	Count
Forest	14000
Grassland	3000

Contribution of Each Location Type (%)

Location_Type	Percentage
Forest	82%
Grassland	18%

Biodiversity Hotspots (Species Richness)

Unique Species Count per Location

Location_Type	Species_Count
Forest	100
Grassland	100

Top Observation Plots

Top 10 Observation Plots

Plot	Count
WOOD-001	200
WOOD-004	120
WOOD-003	110
WOOD-005	100
WOOD-002	80
WOOD-006	70
WOOD-007	60
WOOD-008	60
WOOD-009	60
WOOD-010	60

Habitat vs Location Type

Habitat Comparison Across Locations

Location_Type	Count
Forest	14000
Grassland	3000

Park-Level Activity

Observation Count per Park

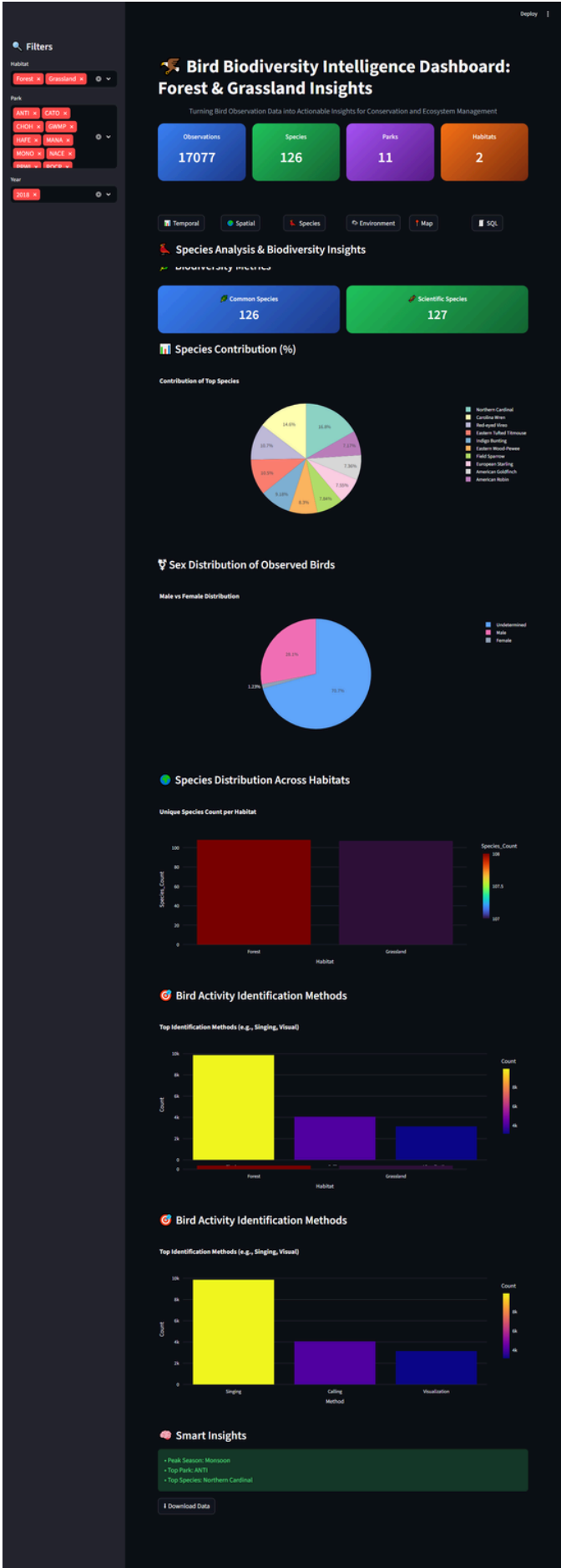
Park	Count
WOOD	4000
PARK	3500
FIELD	2500
WATER	2200
CLIFF	2000
LAKE	1000
RIVER	800
SWAMP	600
BEACH	500
WETLAND	400
DESERT	300

Smart Insights

- Peak Season: Monsoon
- Top Park: WOOD
- Top Species: Northern Cardinal

Download Data

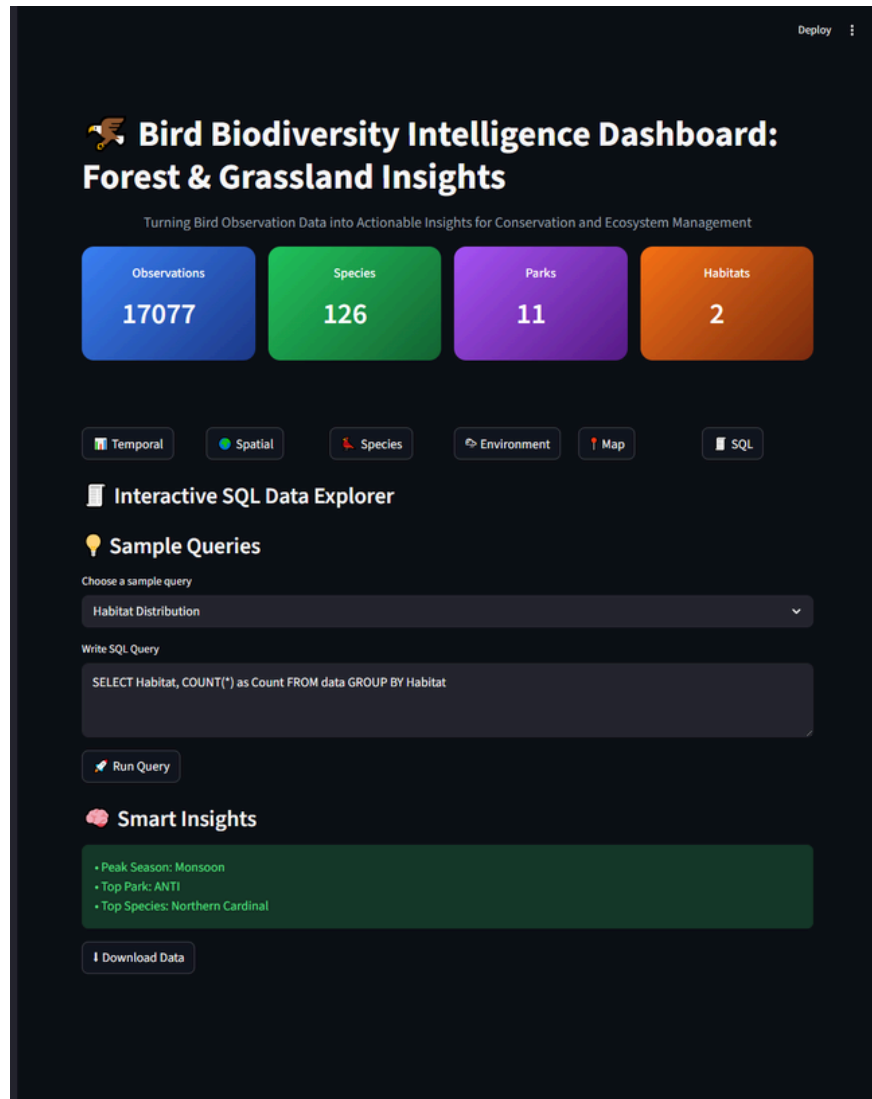
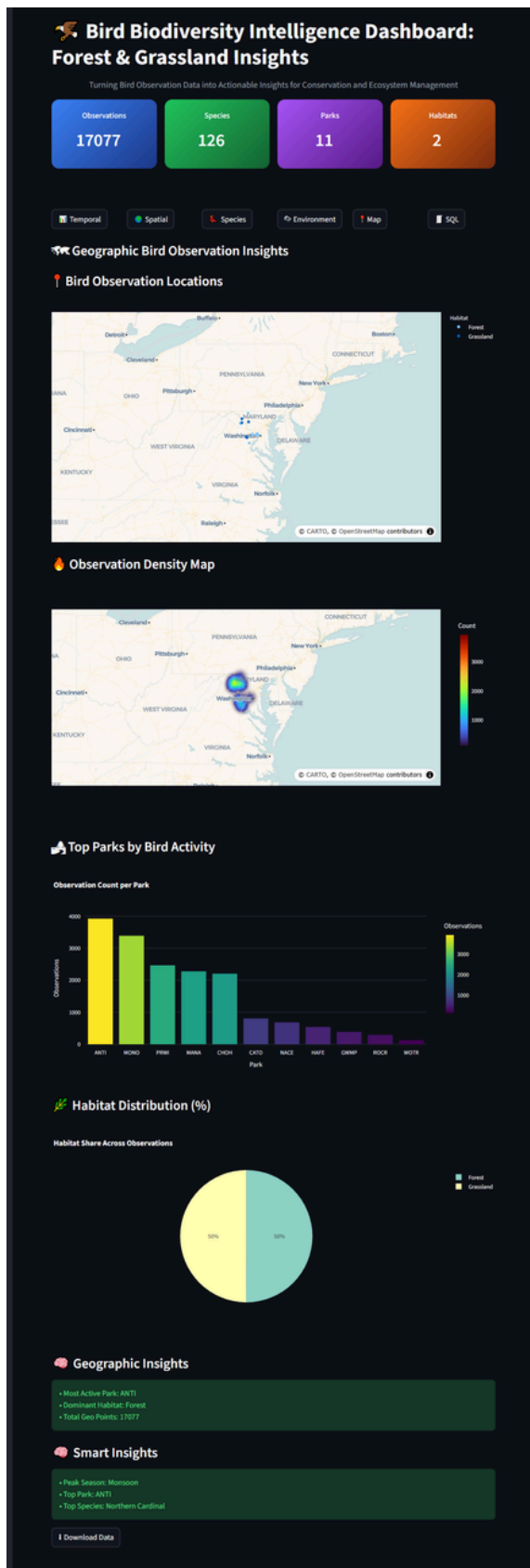
Spatial Dashboard



Species Dashboard



Environment Dashboard



SQL Dashboard

MAP Dashboard

KEY INSIGHTS

6. Key Insights

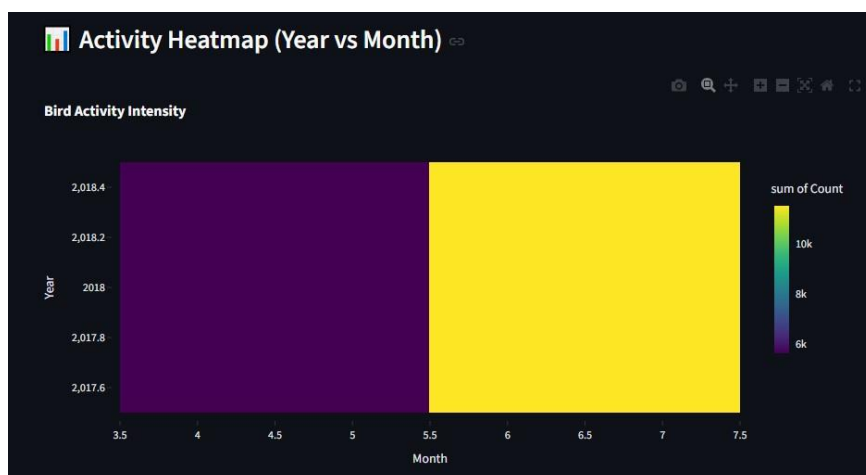
6.1_Temporal Trends:-

- Peak activity during **Monsoon season**
- Lower observations in extreme climates
- Indicates **migration & breeding cycles**

Figure 2: Monthly Bird Activity Trend



Figure 3: Activity Heatmap (Year vs Month)



6.2_ Habitat Analysis:-

Habitat	Observation Share	Biodiversity
Forest	~50%	High
Grassland	~50%	Moderate

🔗 Forests act as **primary biodiversity reservoirs**

6.3_Species Insights:-

- Uneven species observation dominance
- High skewness in distribution
- Implies uneven ecological distribution

Figure 4: Top 10 Most Observed Bird Species

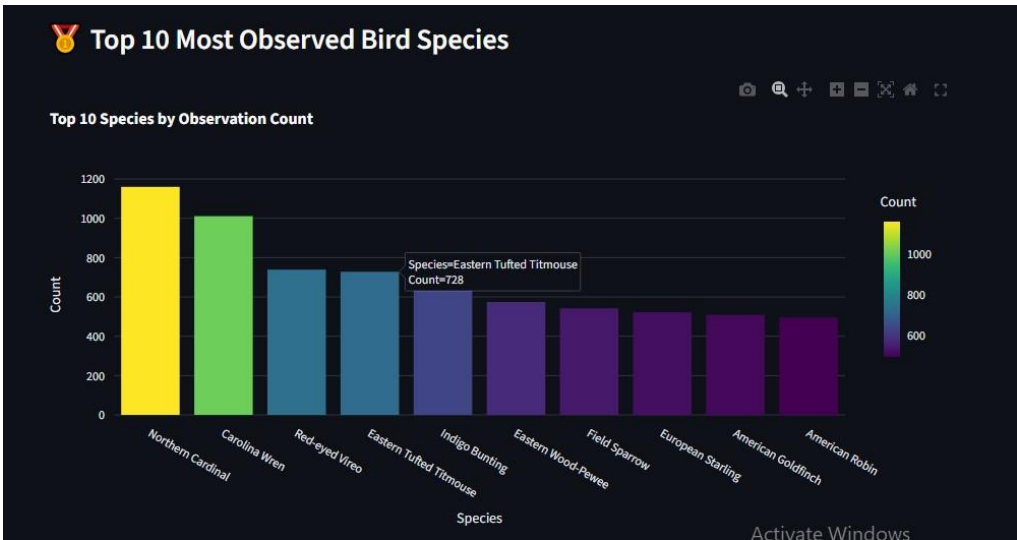
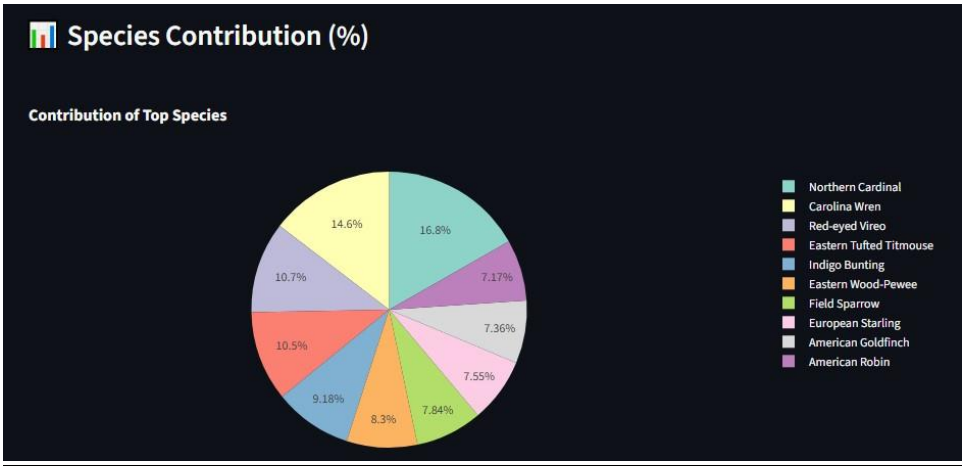


Figure 5: Species Contribution (%)



6.3_Spatial Insights:-

Park	Activity Level
ANTI	Highest
MONO	High
PRWI	Moderate

☞ Identifies biodiversity hotspots

Figure 6: Park-wise Observation Activity

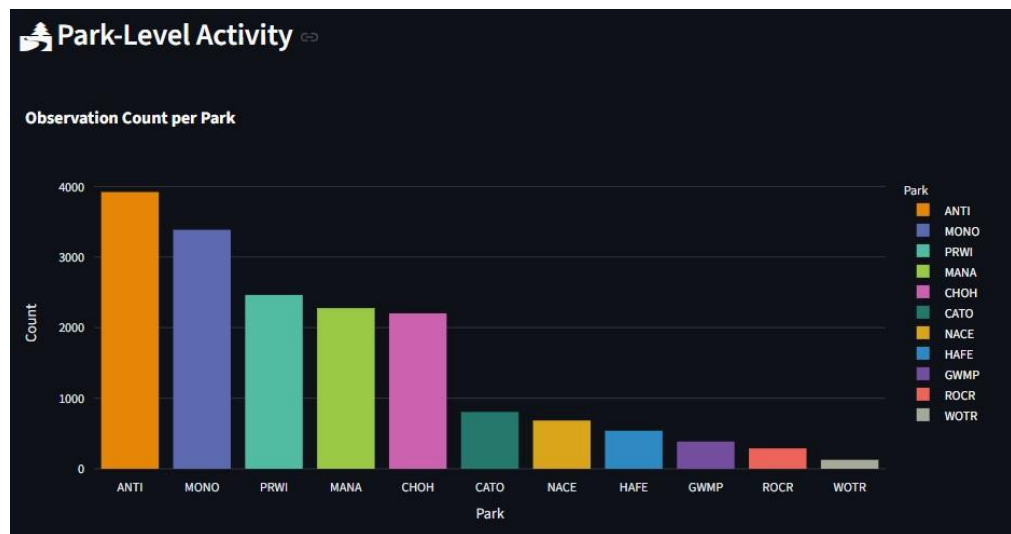
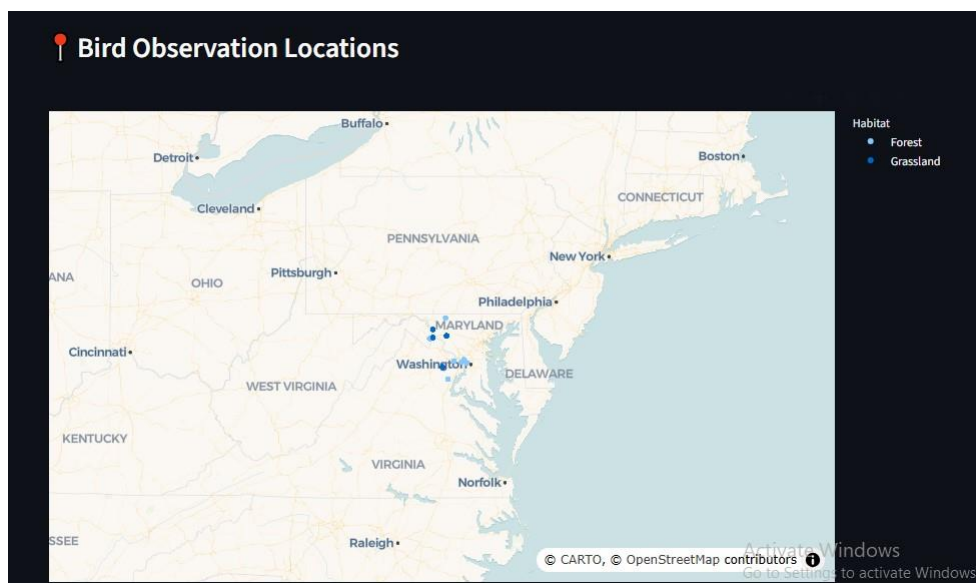


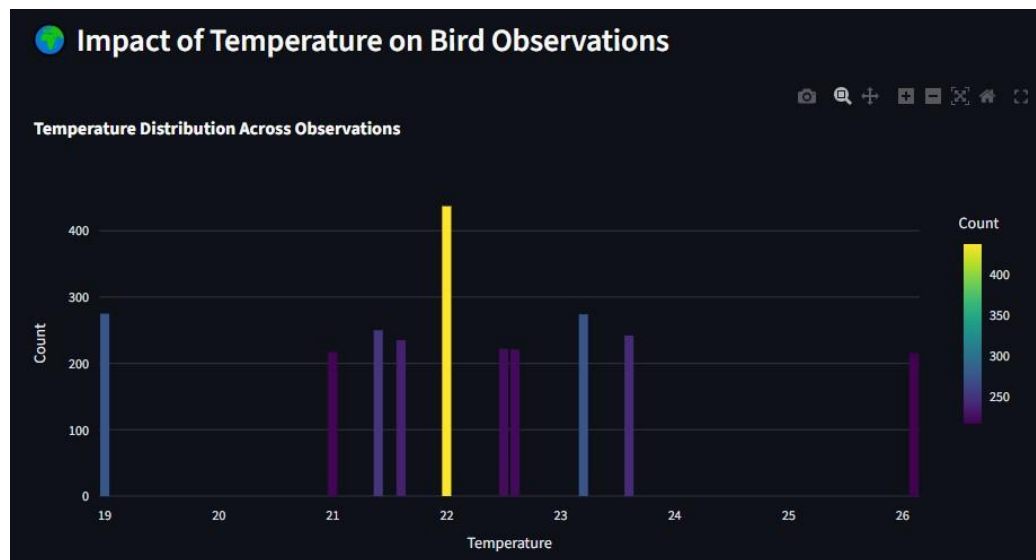
Figure 7: MAP Visualization



6.3_Environmental Insights:-

- Moderate conditions → high activity
- Clear sky → better observations
- Extreme weather → reduced sightings

Figure 8: Environmental Impact



VISUALS & IMPACT

7. Visual Interpretation

Chart Type	Purpose
Bar Chart	Category comparison
Line Chart	Trend analysis
Heatmap	Activity intensity
Pie Chart	Percentage contribution
Map	Geographic distribution

8. Business / Ecological Impact

This analysis supports:

- Conservation planning
- Habitat preservation
- Wildlife monitoring
- Policy decision-making

CHALLENGES & CONCLUSION

9. Challenges Faced

Data Challenges:-

- Missing environmental values
- Inconsistent formats
- No exact GPS coordinates

Technical Challenges:-

- Large dataset handling
- Dashboard performance
- Data transformation issues

10. Conclusion

The dashboard effectively transforms raw data on ecology into useful information.

The important points include:

- Seasonal trends
- Difference in biodiversity based on habitats
- Spatial hotspots
- Role of the environment in observations

This assignment clearly illustrates the utility of data analytics in environmental studies.

FUTURE SCOPE & REFERENCES

11. Future Scope

Though the current dashboard delivers a wealth of information on bird biodiversity in forest and grassland biomes, there are some improvements that would make its analytic capabilities more potent and relevant:

a) Improved Data Integration

- The integration of real-time bird observation and tracking data from third-party APIs or IoT-based tracking systems
- The integration of geospatial datasets (GIS layers) to aid in habitat mapping

b) Predictive and Artificial Intelligence (AI) Analytics:

- The application of machine learning algorithms to predict bird migration and activities
- Species Distribution Modeling (SDM) to predict changes in biodiversity based on environmental conditions

c) Broader Data Scope:

- Additional inclusion of other ecosystems such as wetlands, coastlines, and urban biomes
- Wider scope to include different geographic locations for comparative analysis

d) Enhanced User Interface:

- Use of a mobile-friendly interface design for accessibility purposes
- Adding features for interactive storytelling in the dashboard for non-technical users

e) Conservation Intelligence

- Indicators for bird species' conservation status (IUCN Red List)
- Early warning systems to identify risks in biodiversity

12. References

The analysis and the development of the dashboards rely on the following datasets, tools, and best practices:

Datasets:

- [Forest Birds Monitoring Dataset](#) (Excel Format)
- [Grassland Birds Monitoring Dataset](#) (Excel Format)

Technologies and Tools:

- Python – Data transformation
- Pandas – Data manipulation
- Plotly – Data visualization
- Streamlit – Development of Dashboards

Methods and Best Practices:

- Methods for data cleaning and preprocessing (dealing with missing data, normalization)
- Best practices in exploratory data analysis (EDA)
- Best practices in data visualization